

International Institute of Information Technology, Hyderabad
MA4.101-Real Analysis (Monsoon-2026)

Assignment 2

Total marks: 100

Due date: 6 October, 2026

Note 1

Each of the three number systems $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$ was built in the same way. One writes down a collection of formal expressions, and then *stipulates* when two of these expressions are to be regarded as denoting the same object.

System	Names	Stipulated equality
$\mathbb{Z}$	$a - b, \ a, b \in \mathbb{N}$	$a - b = c - d$ iff $a + d = c + b$
$\mathbb{Q}$	$a // b, \ a, b \in \mathbb{Z}, \ b \neq 0$	$a // b = c // d$ iff $ad = cb$
$\mathbb{R}$	$\text{LIM}_{n \rightarrow \infty} a_n, \ (a_n) \text{ Cauchy in } \mathbb{Q}$	$\text{LIM } a_n = \text{LIM } b_n$ iff $(a_n) \sim (b_n)$

Here $\sim$ is equivalence of sequences. The essential point is that **a name is not the object it names**, and one object has many names.

Definition [Well-defined rule]. A *rule from X to Y* is an assignment R which to each *name* ν of an element of X associates an expression $R(\nu)$. A rule R from X to Y is *well-defined* if both of the following hold.

- (W1) For every name ν of an element of X , the expression $R(\nu)$ is a legitimate name of an element of Y .
- (W2) For all names ν, μ of elements of X , $\nu = \mu \implies R(\nu) = R(\mu)$. This is the substitution property for the rule.

Proposition. Let R be a rule from X to Y . Then R is well-defined if and only if there is a function $f : X \rightarrow Y$ such that $f(x) = R(\nu)$ for every $x \in X$ and every name ν of x and in that case f is unique.

Proof. Suppose R is well-defined. Let $x \in X$. Every element of X has at least one name; choose one, say ν , and set $f(x) = R(\nu)$, which is an element of Y by (W1). This assigns exactly one value to x : if μ is any other name of x then $\nu = \mu$, so $R(\nu) = R(\mu)$ by (W2), and these are the same element of Y . So f is a function with the stated property.

Conversely, suppose such an f exists. Then for any name ν of x we have $R(\nu) = f(x) \in Y$, giving (W1); and if $\nu = \mu$ are names of the same x , then $R(\nu) = f(x) = R(\mu)$, giving (W2).

Uniqueness is immediate: two such functions agree at every x , since each equals $R(\nu)$ for any chosen name ν of x . $\square$

Note 2

Lemma P (componentwise membership). *Let p be an ordered pair, say $p = (x, y)$. Then for all sets X and Y ,*

$$p \in X \times Y \iff (x \in X \text{ and } y \in Y).$$

Moreover the components x and y are determined by p .

Proof. By definition, $p \in X \times Y$ holds if and only if $p = (x', y')$ for some $x' \in X$ and $y' \in Y$. If $x \in X$ and $y \in Y$ we may take $x' := x$ and $y' := y$. Conversely, if $p = (x', y')$ with $x' \in X$, $y' \in Y$, then $(x, y) = (x', y')$, so $x = x'$ and $y = y'$; hence $x \in X$ and $y \in Y$. The same equation, $(x, y) = (x', y') \Leftrightarrow (x = x' \text{ and } y = y')$, gives that the components are determined by p .

Question (1). Let I and J be sets, let A_α be a set for each $\alpha \in I$, and let B_β be a set for each $\beta \in J$.

(a) Prove that

$$\left(\bigcup_{\alpha \in I} A_\alpha \right) \times \left(\bigcup_{\beta \in J} B_\beta \right) = \bigcup_{(\alpha, \beta) \in I \times J} (A_\alpha \times B_\beta).$$

(b) Deduce that for all sets A, B, C, D ,

$$(A \cup C) \times (B \cup D) = (A \times B) \cup (A \times D) \cup (C \times B) \cup (C \times D).$$

(c) Using (b), prove that

$$(A \times B) \cup (C \times D) = (A \cup C) \times (B \cup D)$$

holds *if and only if*

$$(A \subseteq C \text{ or } D \subseteq B) \quad \text{and} \quad (C \subseteq A \text{ or } B \subseteq D).$$

(d) Prove that $(A \times B) \setminus (C \times D) = (A \setminus C) \times (B \setminus D)$ holds if and only if

$$(A \cap C = \emptyset \text{ or } B \subseteq D) \quad \text{and} \quad (A \subseteq C \text{ or } B \cap D = \emptyset).$$

Question (2). Let $f : X \rightarrow Y$ and $g : Z \rightarrow W$ be functions. Define

$$f \times g : X \times Z \rightarrow Y \times W, \quad (f \times g)(x, z) := (f(x), g(z)).$$

(a) Prove that this is a function, and that $(f \times g)(A \times B) = f(A) \times g(B)$ for all $A \subseteq X$ and $B \subseteq Z$.

(b) Prove that $(f \times g)^{-1}(U \times V) = f^{-1}(U) \times g^{-1}(V)$ for all $U \subseteq Y$ and $V \subseteq W$.

Question (3) [Exercise 3.6.6 of Tao]. Let A, B, C be sets. Show that the sets $(A^B)^C$ and $A^{B \times C}$ have equal cardinality by constructing an explicit bijection between the two sets. Conclude that $(a^b)^c = a^{bc}$ for any natural numbers a, b, c .

Question (4) [Exercise 3.6.10 of Tao, The pigeonhole principle]. Let $A_1, \dots, A_n$ be finite sets such that

$$\left| \bigcup_{i \in \{1, \dots, n\}} A_i \right| > n.$$

Show that there exists $i \in \{1, \dots, n\}$ such that $|A_i| \geq 2$. This is known as the pigeonhole principle.

Question (5) [Exercise 4.1.8 of Tao]. Show that the principle of induction (Peano Axiom 5) does not apply directly to the integers. More precisely, give an example of a property $P(n)$ pertaining to an integer n such that $P(0)$ is true, and that $P(n)$ implies $P(n++)$ for all integers n , but that $P(n)$ is not true for all integers n .

Question (6). Answer the following questions.

1. Let integers $\mathbb{Z}$ are defined as collection of names $a \text{ --- } b$, where $a \text{ --- } b = c \text{ --- } d \Leftrightarrow a + d = b + c$ and $a, b, c, d \in \mathbb{N}$. Consider a rule from $\mathbb{Z}$ to $\mathbb{N}$ given by

$$\text{ignore}(a \text{ --- } b) := a,$$

In words: **ignore** assigns to a name its first coordinate. Is **ignore** a well-defined operation?

2. Let rationals $\mathbb{Q}$ are defined as collection of names $a \parallel b$, where $a \parallel b = c \parallel d \Leftrightarrow ad = bc$ and $a, b, c, d \in \mathbb{Z}$ and $b \neq 0$ and $d \neq 0$. Consider a rule from $\mathbb{Q} \times \mathbb{Q}$ to $\mathbb{Q}$ given by

$$\text{mediant}(a \parallel b, c \parallel d) := (a + c) \parallel (b + d).$$

Is **mediant** a well defined operation?

Question (7). Call a sequence (a_n) *rigid* if

there is $N \geq 1$ such that for every $\varepsilon > 0$, $|a_j - a_k| \leq \varepsilon$ for all $j, k \geq N$.

- (a) Prove that (a_n) is rigid if and only if it is *eventually constant*, that is, there is $N \geq 1$ with $a_j = a_N$ for all $j \geq N$.
- (b) Deduce that every rigid sequence is Cauchy, and show that the converse fails.

Question (8). Let $(a_n)_{n=1}^{\infty}$ be a Cauchy sequence of rationals.

- (a) Let φ be any map from $\{n \in \mathbb{N} : n \geq 1\}$ to itself such that $\varphi(n) \geq n$ for every n . Prove that $(a_{\varphi(n)})_{n=1}^{\infty}$ is Cauchy and is equivalent to (a_n) .
- (b) Deduce that (a_{n+1}) , (a_{2n}) and (a_{n^2}) are Cauchy and all name the same real number as (a_n) .
- (c) Show that the hypothesis $\varphi(n) \geq n$ cannot simply be dropped, by exhibiting a Cauchy (a_n) and a map φ for which $(a_{\varphi(n)})$ is Cauchy but not equivalent to (a_n) .